LINUX / SYSTEMS

Linux Filesystem Visual Guide

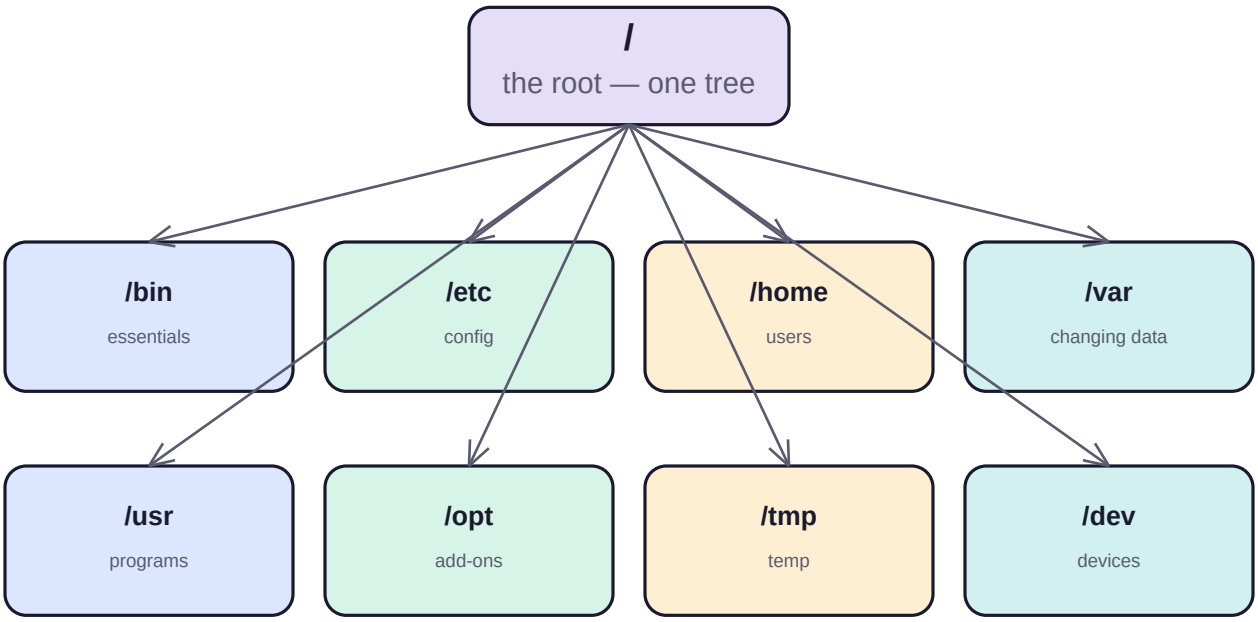
FHS hierarchy, inodes, links, mounts,
and the virtual filesystems /proc and /sys.

ANIMATED WALKTHROUGH
Step through pages to play the animation

STEP 1/6

The FHS Hierarchy

One tree, rooted at /. Everything is a file or a directory.

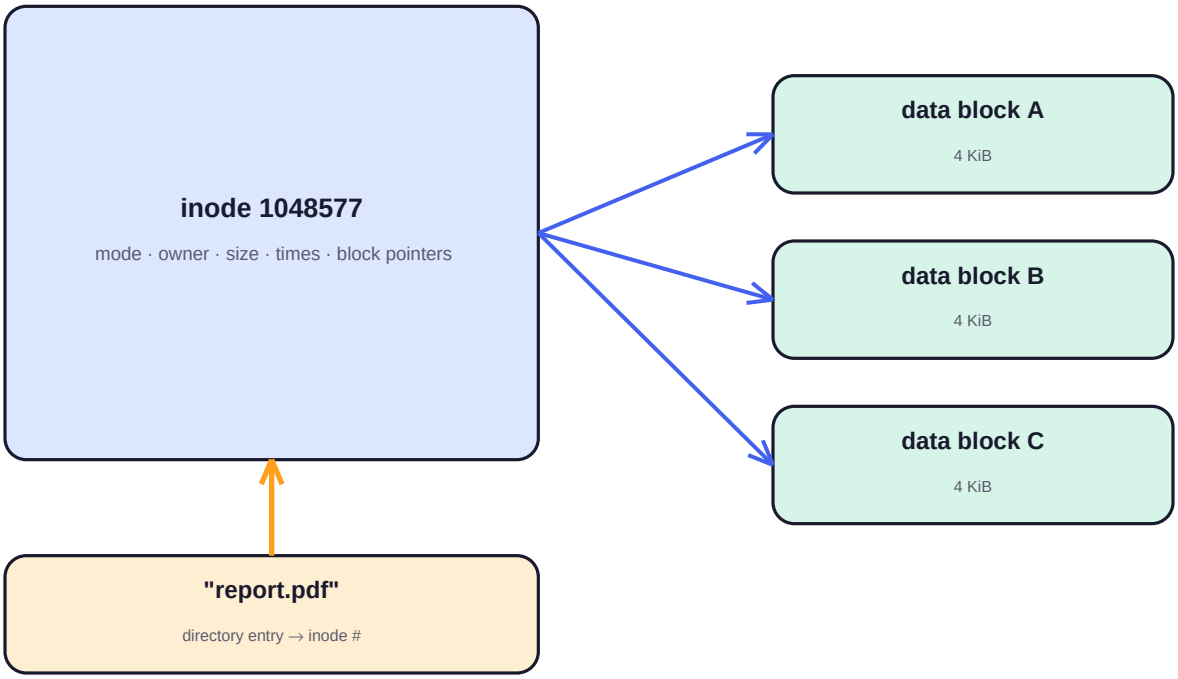


No drive letters: disks are mounted INTO this single tree. /mnt and /media hold removable mounts.

STEP 2/6

Inodes: Files Are Just Numbers

A filename is a label; the inode holds the metadata.

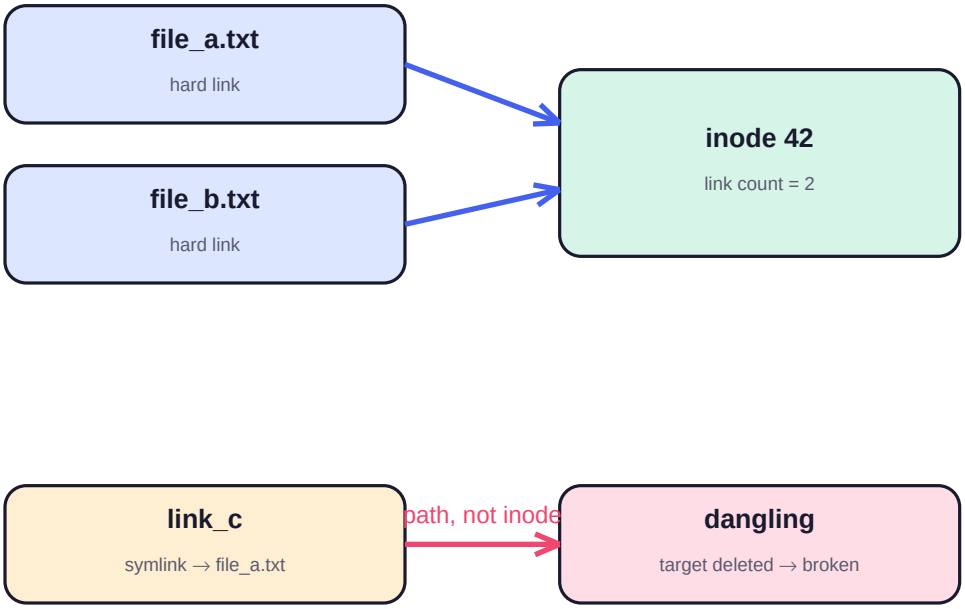


ls -i shows the inode number. Deleting a filename removes the DIRECTORY ENTRY; data dies only when the link count hits zero.

STEP 3/6

Hard Links vs Soft Links

Same data, different relationships.

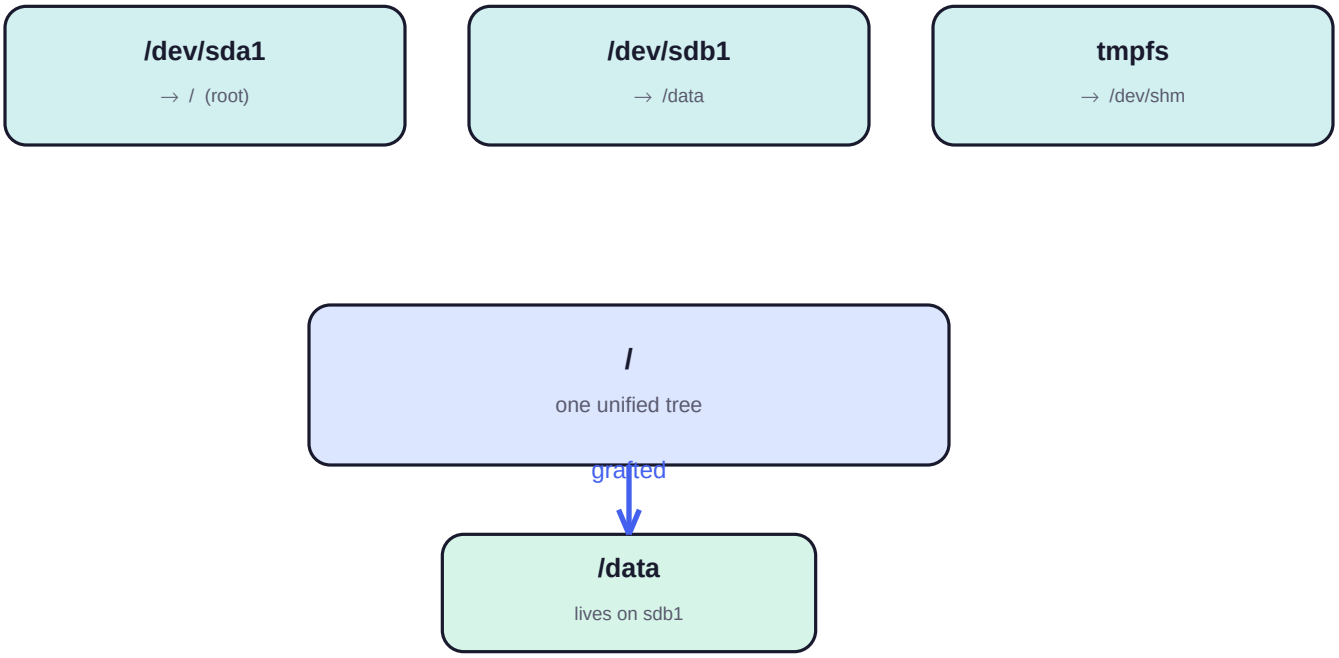


- Hard link: another NAME for the same inode. Same filesystem only. Deleting one name is safe.
- Symlink: a tiny file holding a PATH. Can cross filesystems, can dangle, has its own inode.
- ln file_a file_b (hard) · ln -s file_a link_c (soft)

STEP 4/6

Mounts: Grafting Disks Onto the Tree

mount attaches a device at a directory — the mount point.



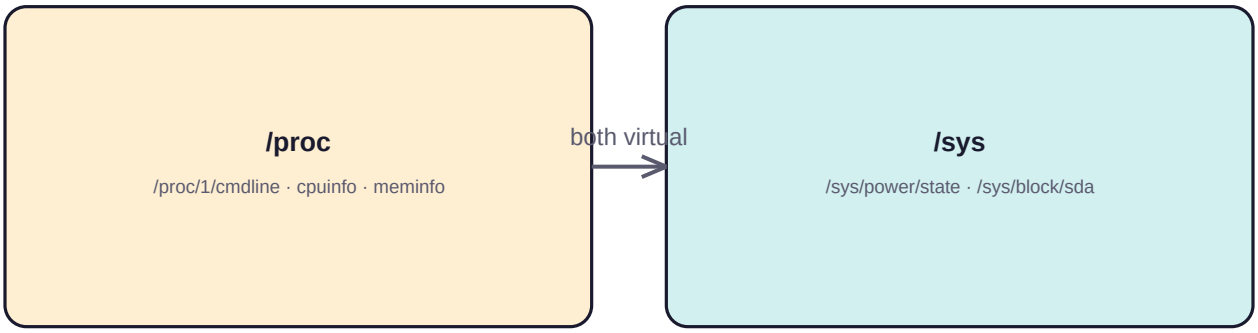

```
$ mount | grep ' /data'
/dev/sdb1 on /data type ext4 (rw,relatime)
$ df -h /data   # space per MOUNT, not per dir
```

Bind mounts (--bind) graft one directory onto another — same data, two paths. /etc/fstab makes mounts survive reboot.

STEP 5/6

/proc and /sys: The Kernel as Files

Pseudo-filesystems: reading a file queries the live kernel.




```
$ cat /proc/cpuinfo | grep 'model name' | head -1
$ cat /proc/meminfo | head -2
$ echo mem | sudo tee /sys/power/state   # suspend!
```

- Nothing on disk: these files are generated on read by the kernel.
- /proc/<pid>/ exposes every process — cmdline, environ, fd/, status.
- /sys exposes devices and tunables; writing a value changes kernel behavior live.

STEP 6/6

Putting It Together: Where Things Live

A quick placement map for daily work.

Path	What goes there	Backed up?
/etc	System configuration	YES
/home	User data	YES
/var/log	Logs	maybe
/var/lib	App state (DBs)	YES
/tmp, /run	Ephemeral	NO
/usr	Programs (read-only-ish)	reinstallable
/opt	Third-party apps	reinstallable

Rule of thumb: separate /home and /var onto their own mounts — a runaway log can't take down the OS.

STEP ✓/✓

Key Takeaways

Linux Filesystem

- One tree rooted at / — disks are grafted on via mounts, never drive letters.
- Inodes hold metadata; filenames are just directory entries pointing at inode numbers.
- Hard links share an inode (same fs); symlinks store a path and can dangle.
- /proc and /sys are live kernel APIs disguised as files — read for insight, write to tune.
- Know where things live: /etc config, /var/log logs, /home users, /tmp ephemeral.

Keep exploring

Pair this guide with the matching video walkthroughs
and hands-on labs at interstitiumlabs.dev